

Aaron Kyle Villegas

Data Analyst | Business Intelligence | SQL, Python | aaronvillegas.vercel.app

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Summary: Applied Mathematics graduate specializing in data analytics, business intelligence, ETL processes, and KPI reporting to uncover actionable retail trends. Highly skilled in translating complex data into business strategies using SQL, Python, and Tableau.

EDUCATION

University of California, Irvine

June 2026

Bachelor of Science, Applied and Computational Mathematics, Specialization in Data Science

GPA: 3.91 | Dean's Honor List | Recipient of Howard Tucker Award in Mathematics | Cum Laude

SKILLS

Languages: Python, SQL, R, C++ | **Libraries:** Pandas, NumPy

Tools: Tableau, PowerBI, BigQuery, Snowflake, AWS S3, Git, PostgreSQL, Excel, REST API, Linux CLI

Methods: Data Visualization, KPI Tracking, ETL Pipelines, Data Cleaning, Statistical Analysis, A/B Testing

RELEVANT EXPERIENCE

Handshake AI – Remote

July 2026

AI Benchmark Engineering Fellow

- Developed robust testing frameworks to identify data anomalies (e.g., time zone shifts, unit glitches), ensuring the integrity of downstream ETL pipelines and executive BI reports.
- Developed an automated Python verification suite utilizing statistical analysis to evaluate data accuracy and maintain strict reporting tolerances.

Jane Street Summer Program Student, Quantitative Finance/Trading | NYC

June 2023 - July 2023

- Selected for competitive quantitative finance program; developed and optimized a Python-based trading strategy with a team in a simulated environment, finishing in the top 25%.

DATA ANALYTICS PROJECTS

Walmart Retail Sales & Inventory Strategy Analysis — Python, SQL, Tableau, Pandas

April 2026

- Developed automated reporting models using Python and SQL to evaluate business KPIs across retail locations, translating raw data into insights that uncovered a 69% holiday sales lift and identified top 20% revenue-driving stores.
- Analyzed store-level demand volatility to track consumer purchasing patterns, recommending flexible inventory and supply chain strategies for volatile locations.
- Designed [Tableau dashboard](#) tracking 9+ core business KPIs (YoY growth, demand variability, 30-day rolling average, store rankings) to directly support executive inventory planning and staffing decisions.

Environmental Health Risk Mapping Platform (Datathon Winner: Best Visualization) ([Live App](#))

April 2026

- Won "Best Visualization" out of 39 teams with an interactive dashboard that predicted asthma and cardiovascular risks.
- Aggregated and cleaned public health data from 8,000+ census tracts (700+ ZIP codes) using Python, engineering a Health Vulnerability Index (HVI) that combined pollution and socioeconomic factors.
- Applied XGBoost machine learning models to identify key predictors of health risks.

A/B Testing Analysis of Digital Marketing Campaigns — Python, Pandas

May 2026

- Executed statistical hypothesis testing (two-proportion z-tests, power analysis) on 588K+ users, confirming a statistically significant 0.7% conversion lift ($p < 0.05$) to guide ad spend allocation.
- Mapped user interaction metrics and conversion funnels in Seaborn/Matplotlib to deliver visual KPIs to marketing leadership.

End-to-end Macro Data Pipeline — Python, PostgreSQL, BigQuery, Snowflake, AWS S3, Linux CLI

April 2026 – Present

- Designed and built an end-to-end ETL pipeline to source time-series data from FRED and Open-Meteo APIs into PostgreSQL, Snowflake, and BigQuery to support large-scale business intelligence initiatives.
- Executed complex SQL queries to uncover historical trends, calculating rolling averages and year-over-year variances to help stakeholders visualize long-term market impacts.
- Automated ETL data-cleaning and deduplication workflows, reducing integration errors and improving the reliability of datasets used for operational reporting.

EMPLOYMENT

Rock Wall Supervisor and Belayer—UCI Recreation Center Irvine, California

April 2024 – June 2026

Student Body Intern – Associated Students at UCI, Irvine, California

Sept 2023 - June 2024